

Example 4.3.2. For the ellipse $9x^2 + 16y^2 = 576$, find the eccentricity e , locate the positions of the foci, focal width, length of major axes and minor axes and equation of directories.

Solution: Put equation in standard form

$$\frac{x^2}{64} + \frac{y^2}{36} = 1$$

Here $a = 8$ and $b = 6$, then the major axis for this ellipse is horizontal.

$$b^2 = a^2 (1 - e^2)$$

$$36 = 64(1 - e^2)$$

$$e = \frac{\sqrt{28}}{8}$$

Then Foci are $(\pm ae, 0) = (\pm\sqrt{28}, 0)$, Focal width $= \frac{2b^2}{a} = 9$, Equation of two directories $x = \pm \frac{32}{\sqrt{7}}$.

Example 4.3.3. Find the equation of ellipse, whose vertex is $(0, 0)$, one of foci is $(0, 3)$ and the length of the major axes is 10

Solution: Since vertex is $(0, 0)$ and one of foci is $(0, 3)$, then the major axis for this ellipse is vertical and $ae = 3$. Since length of major axes = 10, then

$$2a = 10 \Rightarrow a = 5 \Rightarrow e = \frac{3}{5}$$

Since

$$b^2 = a^2 (1 - e^2)$$

$$b^2 = 25 \left(1 - \frac{9}{25}\right)$$

Then $b = 4$ and

$$\frac{x^2}{16} + \frac{y^2}{25} = 1$$

Example 4.3.4. Find the equation, eccentricity e , focal width, the positions of the foci, and equation of directories for an ellipse passes through two points $(4, 3)$, $(6, 2)$.

Solution: Since the points $(4, 3)$, $(6, 2)$ are satisfy the equation of ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \text{ then}$$

$$\frac{16}{a^2} + \frac{9}{b^2} = 1$$

$$\frac{36}{a^2} + \frac{4}{b^2} = 1$$

Hence by solving these equations, we get $a^2 = 52$, $b^2 = 13$ and the equation of ellipse takes the form

$$\frac{x^2}{52} + \frac{y^2}{13} = 1$$

$$13x^2 + 52y^2 = 676$$

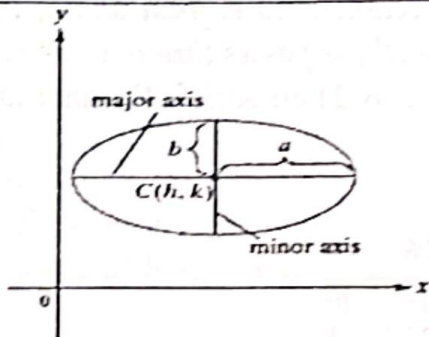
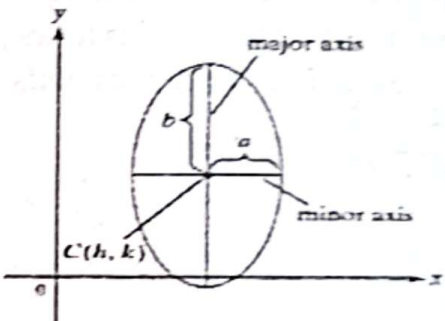
Since

$$b^2 = a^2 (1 - e^2)$$

$$13 = 52 (1 - e^2)$$

Then $e = \frac{\sqrt{3}}{2}$, and Focal width $= \frac{2b^2}{a} = \sqrt{13}$, where $a = 2\sqrt{13}$, $b = \sqrt{13}$ then, the position of foci is $(\pm ae, 0) = (\pm\sqrt{39}, 0)$. Equation of directories $x = \pm \frac{a}{e} = \pm \frac{4\sqrt{13}}{\sqrt{3}}$

The key Point above gives the equation of the ellipse with its center at the origin. If the center of the ellipse has coordinates (h, k) and still has its axes parallel to the x and y -axes, then we have

Key Concept	Equation of ellipse with center (h, k)	
Standard form	$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$	$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$
Major axis	Horizontal	Vertical
Vertices	$(h \pm a, k)$	$(h, k \pm a)$
Foci	$(h \pm ae, k)$	$(h, k \pm ae)$
Equation of directories	$x = h \pm \frac{a}{e}$	$y = k \pm \frac{a}{e}$
Equation of major axis	$y = k$	$x = h$
Equation of minor axis	$x = h$	$y = k$
Graph		

And the general equation of ellipse

$$Ax^2 + Cy^2 + Dx + Ey + F = 0$$

where $A \neq C, AC > 0$

Example 4.3.5. Find the eccentricity e , locate the positions of the foci and center, focal width, length of major axes and minor axes, equation of directories sketch the ellipse

$$4x^2 + 9y^2 - 48x + 72y + 144 = 0$$

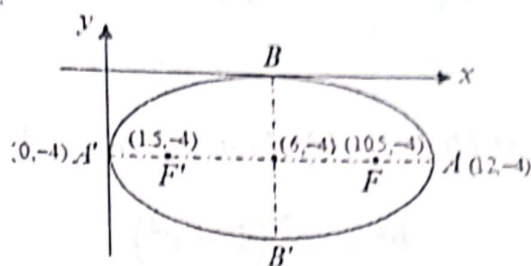
Solution: Complete squares, we get

$$4(x^2 - 12x) + 9(y^2 + 8y) = -144$$

$$4(x - 6)^2 + 9(y + 4)^2 = -144 + 144 + 144$$

$$4(x - 6)^2 + 9(y + 4)^2 = 144$$

$$\frac{(x - 6)^2}{36} + \frac{(y + 4)^2}{16} = 1$$



- Center of ellipse $(h, k) = (6, -4)$, and $a = 6$, $b = 4$, the major axis of ellipse is horizontal.

- Length of major axis = 12, and length of minor axis = 8.

- The vertices are $(h \pm a, k)$, then the vertices are $(0, -4)$ and $(12, -4)$.

- $b^2 = a^2(1 - e^2) \Rightarrow e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{\sqrt{5}}{3}$.

- Location of foci are $(h \pm ae, k)$, $= (6 \pm 2\sqrt{5}, -4) \approx (10.5, -4)$ and $(1.5, -4)$

- Equation of directories is $x = h \pm a/e = 6 \pm 18/\sqrt{5}$